

**ATTACHMENT 31 – CONTRACTOR TRANSMITTAL (SUBMITTAL NUMBER
723.480-008 D) FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED
REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED
FEBRUARY 4, 2000 REGARDING THE INSTALLATION PROCEDURE OF THE
ADHESIVE ANCHORING SYSTEM AND OMITTING THE USE OF SCREEN
TUBES
(6 pages)**



Central Artery/Tunnel

OFFICE COPY

CONTRACTOR TRANSMITTAL

TRANSMITTAL DATE: 12/30/89

TRANSMITTAL NO: 07-345

CONTRACTOR:

Robert S. Smith

ATTENTION:

Robert S. Smith

CONTRACT NO:

99119-00932

DOCUMENT TYPE

- R. REPRODUCIBLE
- P. PRINT
- C. CATALOG CUT
- E. EQUIPMENT MANUAL
- S. SAMPLE
- T. TECHNICAL DATA OR CALCULATIONS

REVIEW STATUS

- 1. WORK MAY PROCEED
- 2N. WORK MAY PROCEED SUBJECT TO THE CHANGES INDICATED. RESUBMITTAL NOT REQUIRED.
- 2R. WORK MAY NOT PROCEED ON INDICATED PART(S). RESUBMITTAL REQUIRED. WORK MAY PROCEED ON OTHER PART(S).
- 3. WORK MAY NOT PROCEED. REVISE AND RESUBMIT.
- 4. FOR INFORMATION ONLY.

DATE RECEIVED BY BIPB RE OFFICE: 12/30/89

DATE DUE BACK TO CONTRACTOR: 1/7/90

DATE SENT BACK TO CONTRACTOR: 2-4-90

RESIDENT ENGINEER:

DATE: 2-4-90

ITEM	SUBMITTAL NO. (I.E., 120.010-011-A)			DESCRIPTION	DOC DATE	DOC TYPE	REVIEW STATUS
	SPEC SECT	- SEQ#	- REV				
1	123-180	008	D	Adhesive Anchors for Concrete Surfaces		T	ZN
2							
3							
4				Review of Construction with Surveys			
5				No. 123-180 - side "Structural Calculations"			
6				for Adhesive Anchors			
7							
COMMENTS TO CONTRACTOR (USE ADDITIONAL SHEETS AS NECESSARY)							
Provide Final of the attached Data RPT E-7517 and include resp. to reviewer comment on section 2.4.4							
REVIEWER:					DATE: <u>2-4-90</u>		



Adhesive Anchors for Ceiling Supports

INDEX STAMP

Contract No. 99119-C09B2	Paragraph 2.08
Specification Section (Spec-Seq-Rev)	
723.480-008 D	

CENTRAL ARTERY / TUNNEL	
Permission To Proceed Does Not Constitute Acceptance Or Approval Of Design Details, Calculations, Analysis, Test Methods, Or Materials Developed Or Selected By The Contractor / Supplier And Does Not Relieve The Contractor / Supplier From Full Compliance With Contract Requirements.	
☐	1. Work May Proceed
☒	2N. Work May Proceed Subject To The Changes Indicated Resubmittal Not Required
☐	2R. Work May Not Proceed On Indicated Parts Resubmittal Required. Work May Proceed on Other Parts.
☐	3. Revise and Resubmit. Work May Not Proceed.
☐	4. For Information Only.
REVIEWER	
<i>Gannett & Henry</i> 2-4-00	
	Company Date

K:\9809\SUBMIT\SUBSTAMP.WK4

Provide final of the attached
Draft Rpt ER-4514 and include reviewer
comment on 2.4.4.
response to

Precast Concrete Ceiling Adhesive Anchoring System
Submittal No. 723.480 - 008

The intent of this submittal is to provide all relevant Quality Control and material information for the provision and installation of the adhesive anchoring system required for the installation of the precast ceiling modules and other structural components, as described in Specification Section 723.480, Sections 1.04N and 2.08. The attached information from Power's Fasteners, Inc. (formally "The Rawlplug Company") shall be applicable to the anchors indicated in the contract documents, including but not limited to the following:

<u>Anchor Size</u>	<u>Sheet</u>	<u>Detail</u>
5/8" Diameter	S-279 S-387	Detail 2 / Section C Section A,B,C,D, Detail 2 and Detail – Vertical Hanger Support

Manufacturer's information on the above listed anchor sizes has been provided in Section 1 of the submittal package. Relevant information on all anchoring components, including but not limited to epoxy adhesive, dosage control screens, plugs, anchor rod, nuts, washers and locking devices has been included. Test results from the anchor test as described in Section 723.480, Section 2.08 B.1. have also been included for review.

Design Calculations reflecting the use of stainless steel anchors and Samples of the Adhesive Anchor System shall be submitted under separate cover.

A response to the Engineer's comments has been provided in Section 1 for review and consideration by the Project. Standard product information has been provided for review in Section 2. A Quality Control Program and installation procedure associated with the installation of the adhesive anchoring system has been included in Section 3.

Section 3

Installation Procedure & Quality Control Program

Adhesive Anchoring System

Power's Fasteners, Inc.

Section 3 - Installation Procedure & Quality Control Program **Adhesive Anchoring System**

It is the intent of Modern Continental Construction to perform the installation of the adhesive anchoring system in accordance with the included manufacturer's product information, and to utilize the accessories and equipment for epoxy application as supplied by the manufacturer to ensure proper installation of the adhesive anchor system. Modern Continental Construction shall make provisions to provide the proper temperature of adhesive materials and substrate during cold weather applications, per the manufacturer's recommendations.

The following detailed sequence of anchor installation is proposed in accordance with the Manufacturer's recommendations included in Section 1 of this submittal package.

1. Locations for the proposed structural element (clip, angle, etc.) shall be determined by survey and indicated on the substrate. The survey shall indicate the proper orientation, alignment and elevation (if applicable) of each structural element.
2. A template (or the actual structural element, if available) shall be used to determine the proposed location of each anchor required. The proposed location of these anchors shall be indicated on the substrate.
3. The holes shall be drilled in the proper locations. A template shall be used to ensure the drill and drill bit are properly oriented to the substrate while drilling the hole. A drill stop shall be used to ensure proper depth of drill hole. Drill hole diameter shall be determined in accordance with manufacturer's recommendations. MCC shall confirm that the employees performing the work are equipped with proper sizes of drill bits, anchors and accessories for the proposed installation. A catalog cut of the proposed drilling equipment has been included for review in Attachment A.
4. The drill hole shall be brushed clean, free of drilling debris and dust. The anchor will be inserted into the hole to ensure proper sizing of the drill hole.
5. On floor applications, the drill hole shall be filled approximately one half full with the epoxy adhesive, starting from the rear of the hole.

On wall and ceiling applications, after the hole is drilled and cleaned, insert the proper size anchor seal plug into the hole until the rim of the plug is fully seated against the base material. Insert the mixing nozzle through opening in the seal plug, and fill the drill hole approximately one half full, starting from the rear of the hole.

6. The anchor shall be cleaned free of dirt, oil or foreign matter and inserted into the hole (through the anchor seal plug in wall and ceiling applications), turning the anchor to ensure positive disbursement of the epoxy. The anchor shall be fully inserted to the proper embedment.
7. The anchor shall not be disturbed or loaded until after the minimum specified cure time for the epoxy, per the manufacturer's recommendations.